

Case Study 1: Credit Risk & Loan Portfolio Analysis

Difficulty: (Hard)

Business Scenario

A bank has **2 million loan records**. The Risk Management team wants to identify high-risk customers and calculate portfolio risk metrics.

Input Files

customers.csv

loans.csv

credit_score.csv

Requirements

Python

- Read multiple CSV files
- Exception handling for corrupted files
- Functions
- OOP (Loan class)

NumPy

Calculate

- ✓ Mean Loan Amount
 - ✓ Median Salary
 - ✓ Percentile Interest Rate
 - ✓ Correlation between Salary and Loan Amount
 - ✓ Standard Deviation
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Pandas

Merge

- ✓ customers
 - ✓ loans
 - ✓ credit_score
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Find

- ✓ Top 20 risky customers
 - ✓ Customers having
 - ✓ Credit Score <650
 - ✓ Salary <60000
 - ✓ Loan >10 Lakhs
 - ✓ DefaultFlag=1
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Missing Data

Replace

- ✓ Salary → Median
 - ✓ Credit Score → Mean
 - ✓ Interest Rate → Previous Value
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Outliers

Remove

- ✓ Loan Amount >99 percentile

Finance Metrics

Calculate

- ✓ Debt-to-Income Ratio
- ✓ Loan Utilization
- ✓ Default %
- ✓ NPA %
- ✓ Average EMI
- ✓ Expected Loss

Automation

Generate

- ✓ risk_report.xlsx
 - ✓ high_risk_customers.csv
 - ✓ summary.json
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